Basic Questions (Any 8 out of 9)

1. How many unique students are there in the dataset?
   Answer: There are total 4894 unique students present in the dataset.

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Total Students: 4894
```

2. What is the average GPA of the students?
   Answer : Average GPA of students is 8.04.

```
Average GPA: 8.04
```

3. What is the distribution of students across different graduation years?
   Answer:
   - 1.53k in 2023
   - 1.51k in 2024 • 1.29k in 2025
   - 0.56k in 2026

Year of Graduation	count
2023	1536
2024	1511
2025	1292
2026	555

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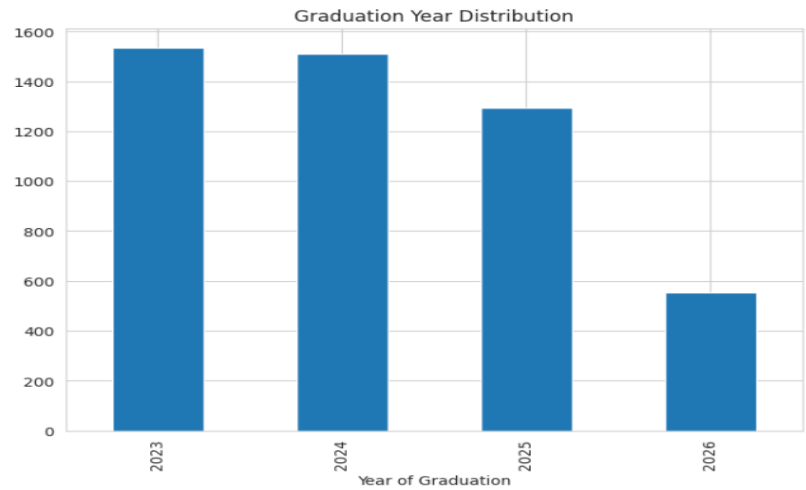

4. What is the distribution of student's experience with python programming? Answer:
   - More than 1000 students have both 3 and 5 months of experience which is the highest count.
   - Over 700 students have 6 months experience.
   - Over 400 students have experience of 4 months which is the lowest count.

Experience with python (Months)	
3	1008
4	466
5	1242
6	738
7	640
8	800

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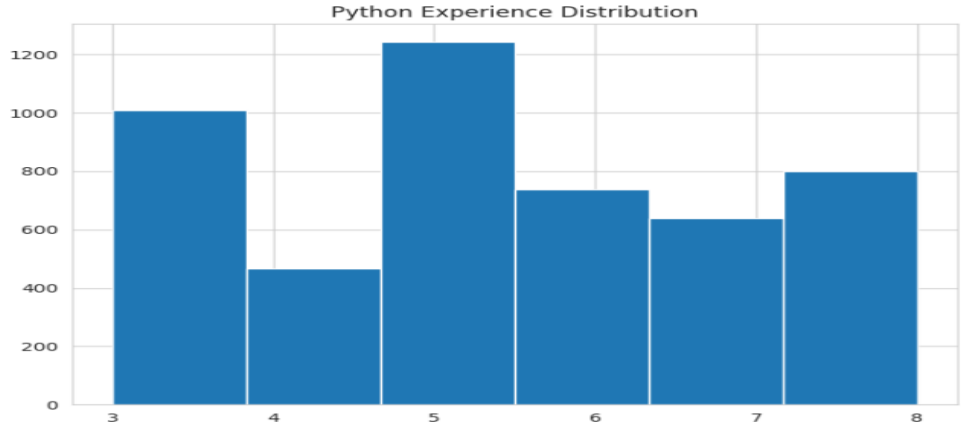

5. What is the average family income of the student?
   Answer : Average family income of the student is 0-2 lakhs as 4.5k students have belong to this range of income.

Family Income	
0-2 Lakh	4525
2-5 Lakh	200
7 Lakh+	89
5-7 Lakh	80

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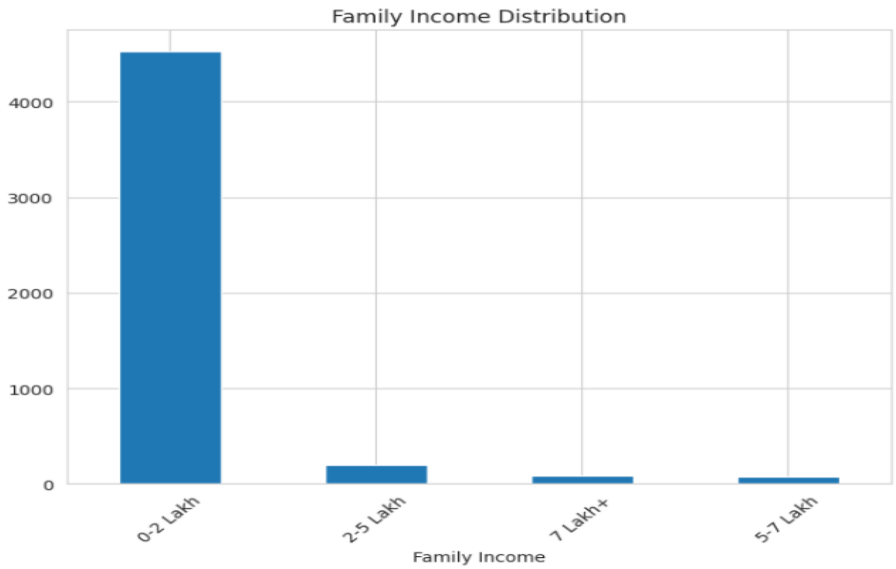

6. How does GPA vary among different colleges? (Show top 5 results only) Answer:
   1. St.Xavier's College – 8.57
   2. Thakur Institute -8.58
   3. AP Shah Institute -8.28
   4. B.K. birla College -8.45
   5. Symboisis Institute -8.30

College Name	
THAKUR INSTITUTE OF MANAGEMENT STUDIES, CAREER DEVELOPMENT & RESEARCH - [TIMSCDR]	8.586
St Xavier's College	8.579
B. K. Birla College of Arts, Science & Commerce (Autonomous), Kalyan	8.456
Symbiosis Institute of Technology, Pune	8.303
AP SHAH INSTITUTE OF TECHNOLOGY	8.283

dtype: float64

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  plt.tight_layout()
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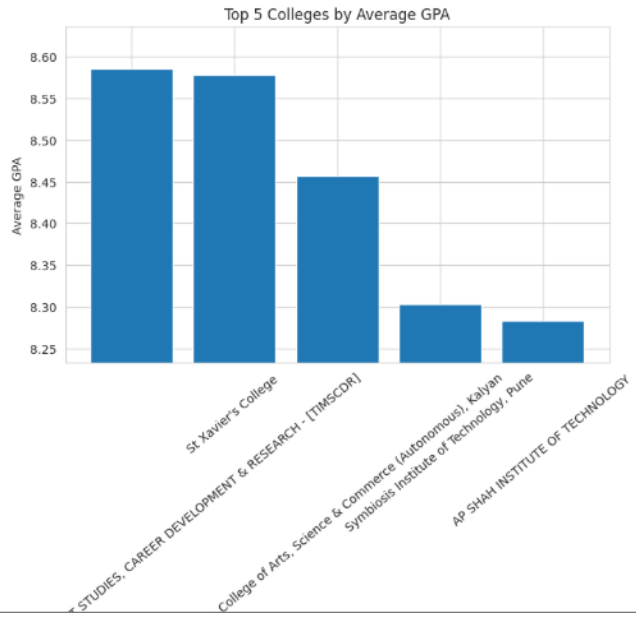
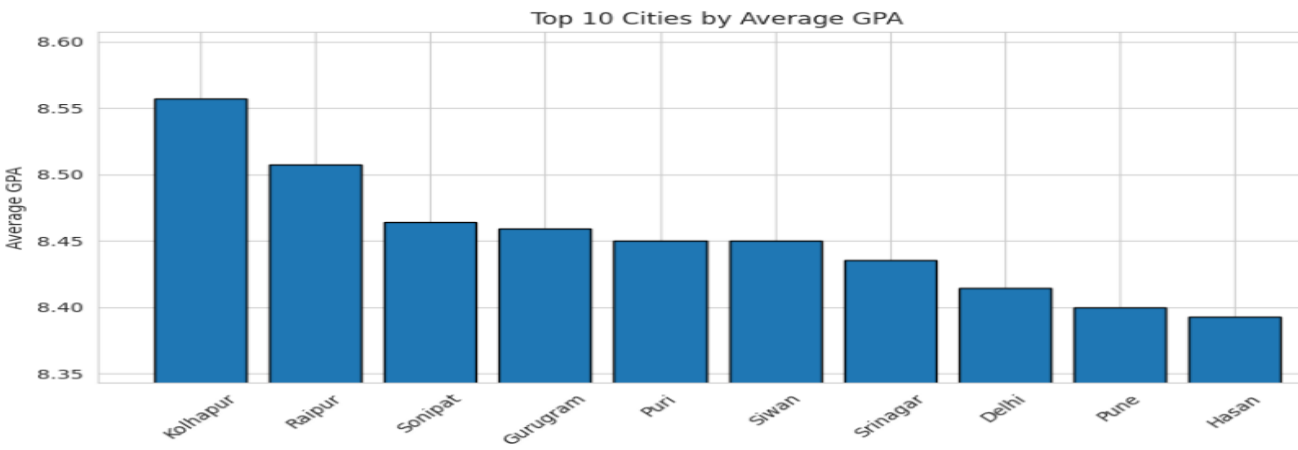


7. What is average GPA for student from each city ?

Answer: Kolhapur has highest GPA followed by Raipur, Sonipat,Gurugram ,Puri and Siwan.



8.Can we identify any relationship between family income and GPA?

Answer:

- Students with a family income of "0-2 Lakh" have an average GPA of around 8.034.
- Students with a family income of "2-5 Lakh" have a slightly higher average GPA of approximately 8.115.
- Students with a family income of "5-7 Lakh" have a lower average GPA of about 8.004.
- Students with a family income of "7 Lakh+" have the highest average GPA of around 8.149.

There seems to be an inverse relationship between family income and average GPA, where students from higher income families tend to have slightly lower GPAs compared to those from lower income families.

Family Income	
0-2 Lakh	8.034
2-5 Lakh	8.115
5-7 Lakh	8.004
7 Lakh+	8.149

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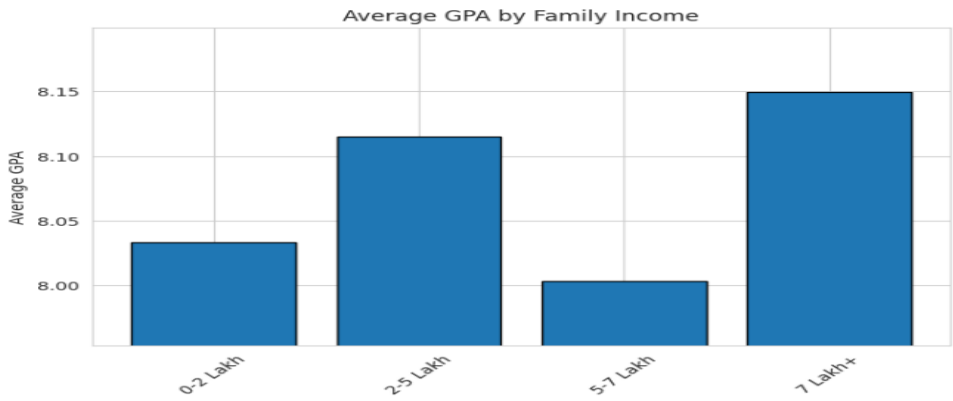

Moderate Questions (Any 8 out of 9)

1. How many students from various cities?

Answer: Chandigarh has highest student count followed by Gonda,Talmuk,Siuri and Jodhpur

City	
Chandigarh	57
Siuri	56
Gonda	56
Talmuk	56
Dwarka	52
Bikaner	52
Amreli	52
Ahemdabad	52
Jaipur	52
Sikar	52

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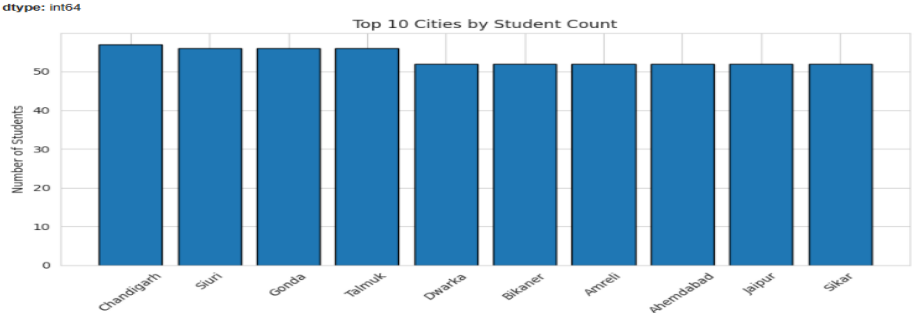

2. How does the expected salary varies based on factor like GPA, Family income, Experience in python (Months)?

Answer: In this GPA Expected Salary graph it is observed that GPA and Expected salary of applicants are proportional , applicants with more GPA expects more salary.

GPA_Range	
<7	13.038
7-8	13.166
8-9	12.914
9+	16.648

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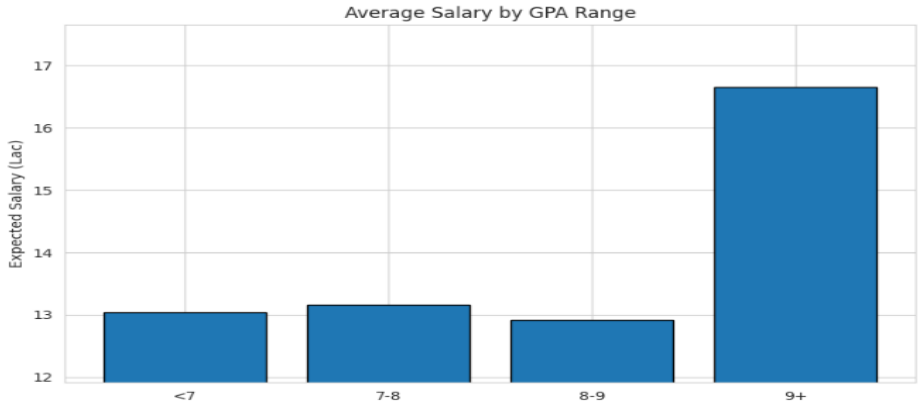

Family Income	
0-2 Lakh	13.874
2-5 Lakh	14.320
5-7 Lakh	15.688
7 Lakh+	14.607

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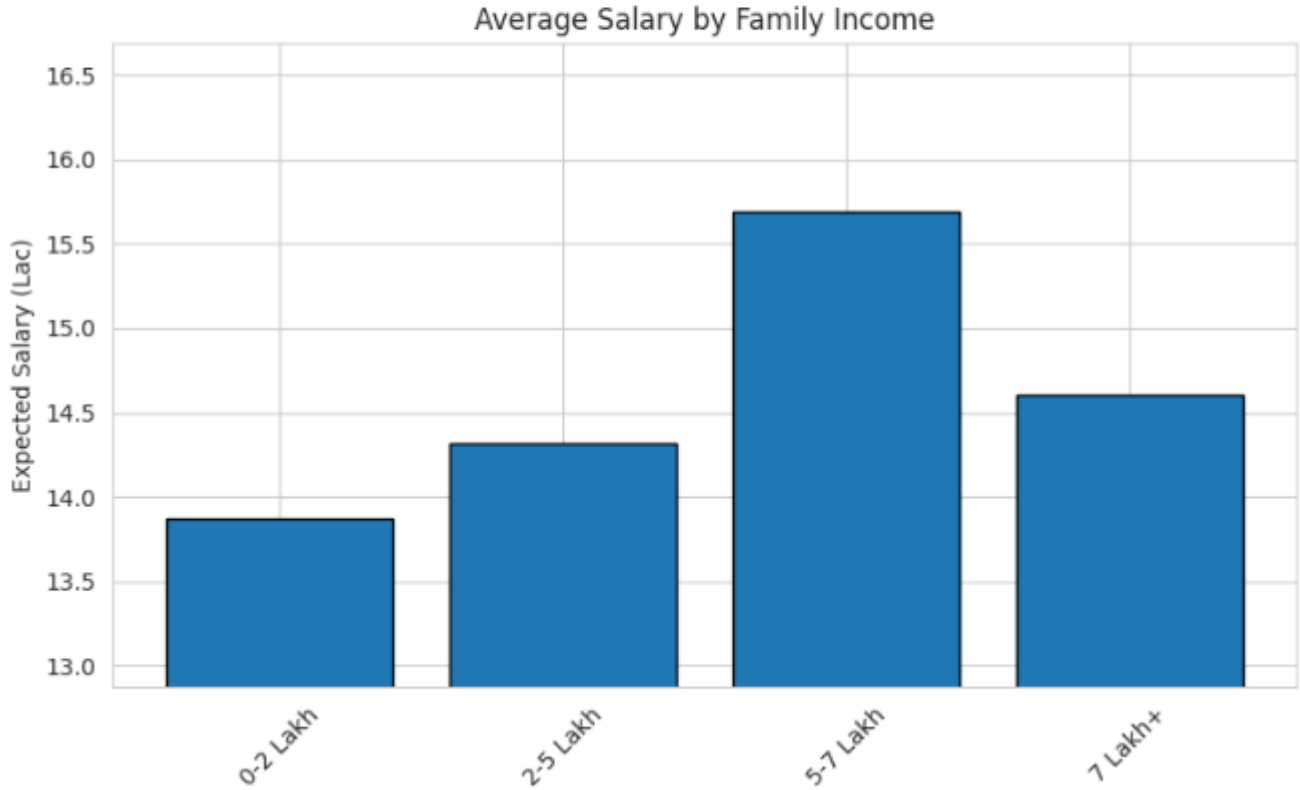

In this income vs expected salary graph it is observed that people between 0-5 lakh average income expect salary ranging from 13-14 LPA, and people between 5-7 lakh average income expect salary ranging from 14-15 LPA.

In this expected salary vs experience in python (months) graph it is analysed that people with more experience expect more salary.

Experience with python (Months)	
3	13.192
4	13.393
5	12.994
6	14.835
7	15.055
8	14.926

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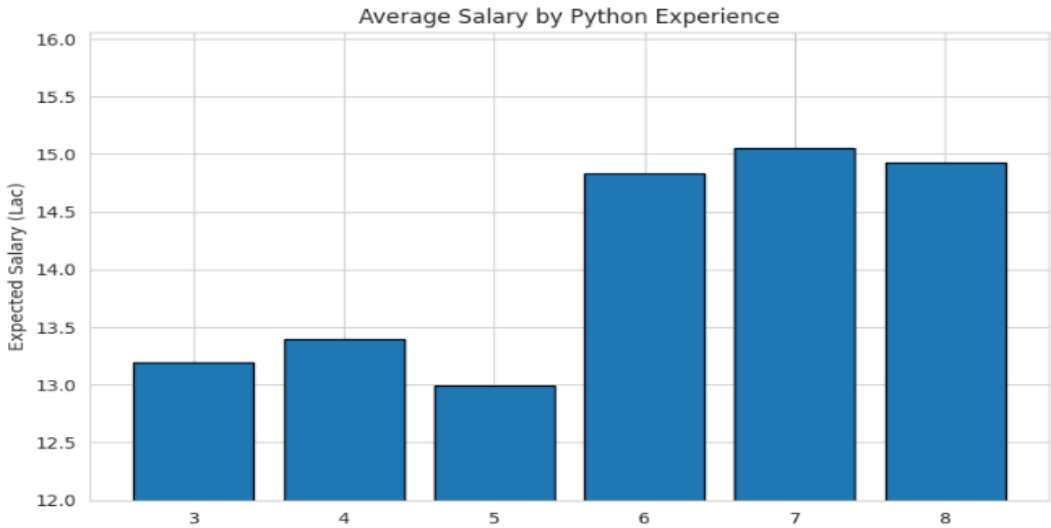
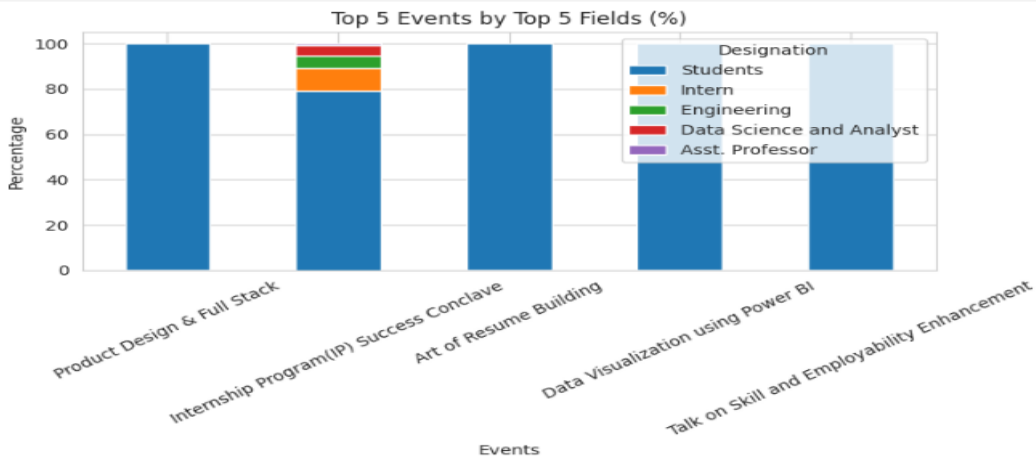


3. Which event tend to attract more students from specific fields of study?

Answer:

Designation Events	Students	Intern	Engineering	Data Science and Analyst	Asst. Professor
Product Design & Full Stack	842	0	0	0	0
Internship Program(IP) Success Conclave	545	70	35	32	6
Art of Resume Building	478	0	0	0	0
Data Visualization using Power BI	455	0	0	0	0
Talk on Skill and Employability Enhancement	379	0	0	0	0

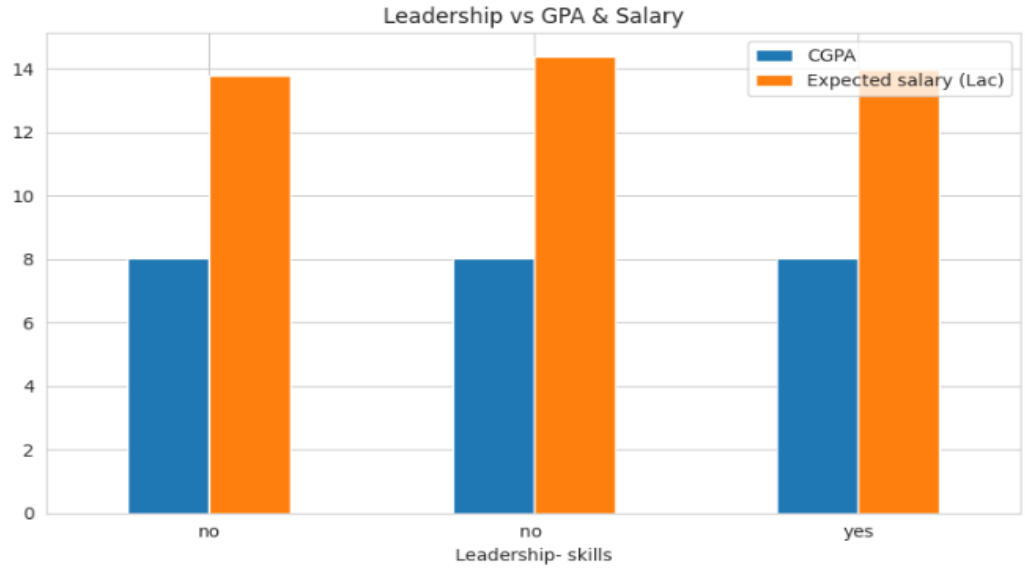


Events like Product design & full stack , IP Success Conclave attract more students with maximum count of 842 applicants attended Product design & full stack event followed by other events and mostly the attendees are students.

4. Do students in leadership positions during their college years tend to have higher GPAs or better expected salary?

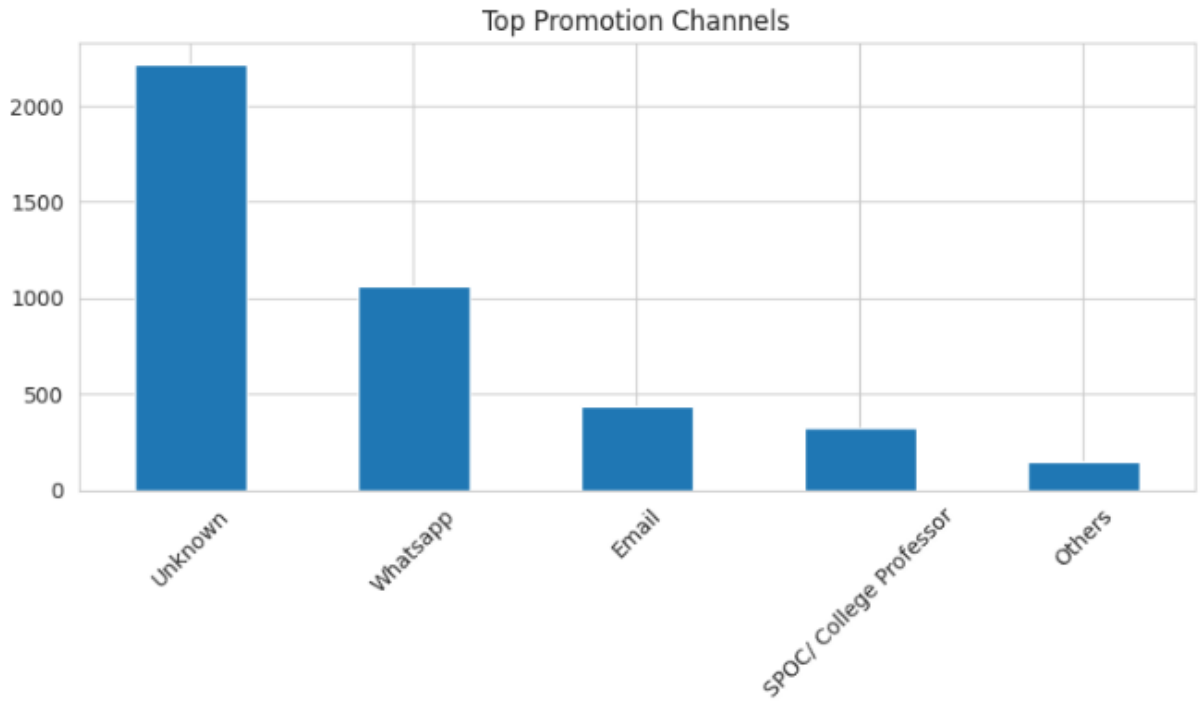
Answer: It is found that participating in leadership positions does not affect their GPAs but people with leadership experience expect better salary than those who did not.

Leadership- skills	CGPA	Expected salary (Lac)
no	8.040	13.795
no	8.035	14.408
yes	8.037	13.972



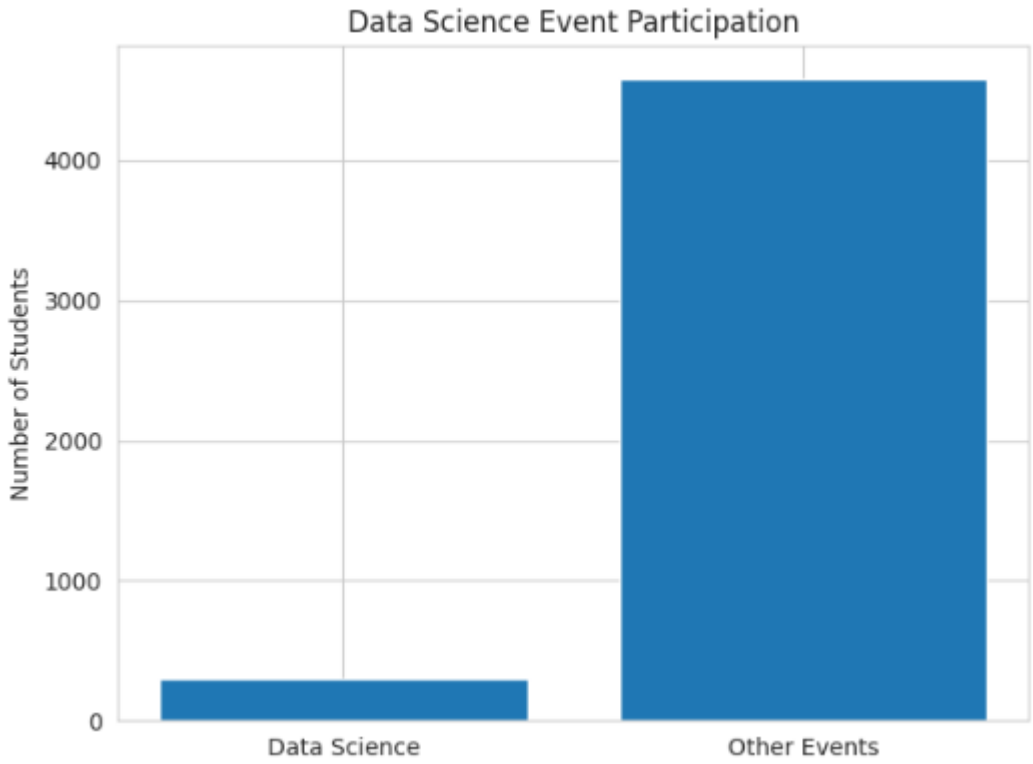
5. What promotion channel brings in more students participants in event?

Answer: Keeping blank aside, Whatsapp bring most number of participants for the event.



6.Find the total number of students who attended the events related to Data Science?

Answer: Total 306 students participated in events related to Data Science.



7.Those who have high cgpa and more experience in language those who had high expectations for salary? (AVG)

	0
High Performers Avg Salary	16.920
Overall Avg Salary	13.936

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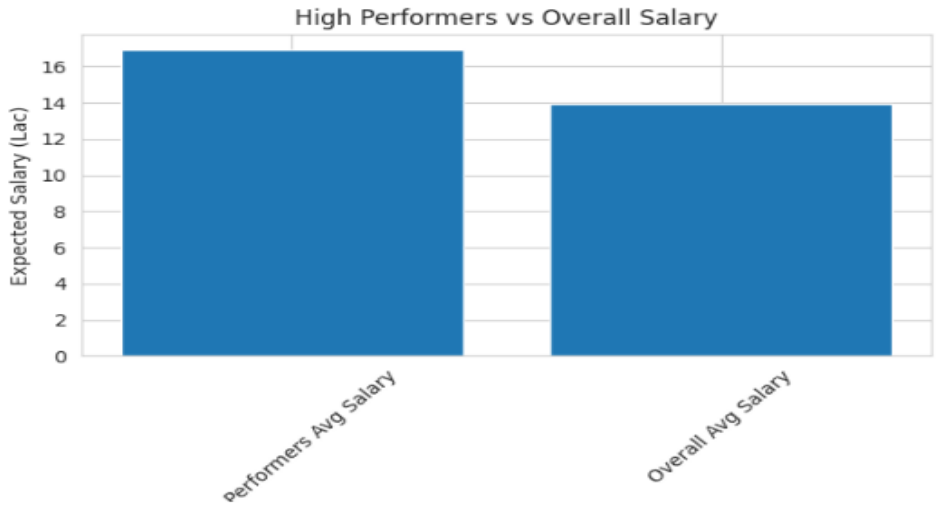
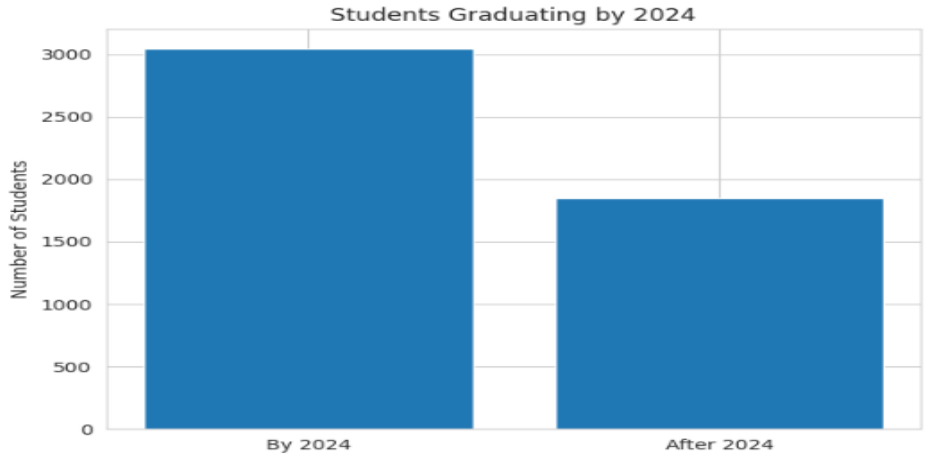


8.How many Students will be gruaduated by 2024 ??

Answer : By 2024 3047 students will be graduated and after 2024 1847 students will be graduated.

Year of Graduation	
True	3047
False	1847

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